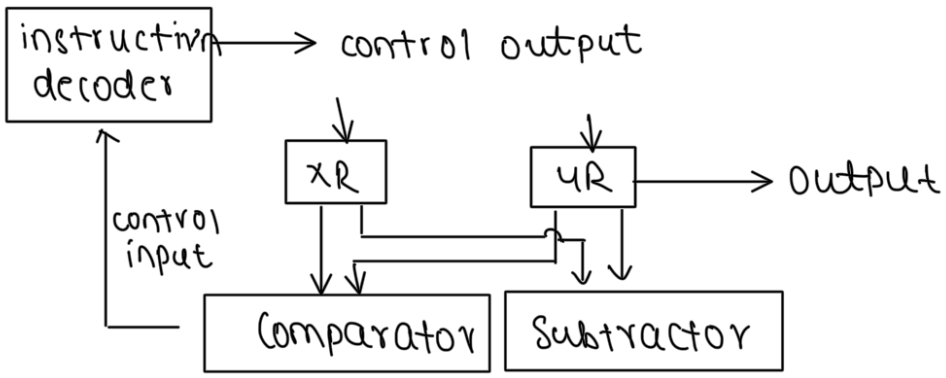
(i)



Case A

T_{REQ} → initiating peripherals operations

T_{CHK} → checking it op^n is done/not

$T_{I/O}$ → time to do job $= N_1 T_{INS}$

T_{RES} → write result

Case B

T_{REQ}

$T_{POLL} = N_2 T_{INS} + T_{CHK}$ (polling interval)

Case C

T_{REQ}

$T_{I/O}$

T_{RES}

$T_{CHK} = T_{INS}/3 = 2T_{CLK}$

$T_{INS} = 6T_{CLK}$

1 I/O ← 100 instructs (T_{INS})

$T_{RES} = T_{INS}/2$

$T_{REQ} = 10T_{INS}$

(a)

Case A:

$(T_{REQ} + T_{I/O} + T_{RES}) + 99 \times 6T_{CLK}$

$= 60T_{CLK} + N_1 \cdot 6T_{CLK} + 3T_{CLK} + 594T_{CLK}$

$= (657 + 6N_1)T_{CLK}$

Case C: $99T_{INS} + T_{REQ} + T_{RES}$ (T_{110}) in parallel

$= 657T_{CLK}$

Case B:

$T_{33} = (63 + 6N_1)T_{CLK}$

no of. polls $= \left\lfloor \frac{N_1 T_{INS}}{N_2 T_{INS} + T_{CHK}} \right\rfloor \rightarrow \left\lfloor \frac{T_{lb}}{T_{poll}} \right\rfloor$

instructions executed concurrently during 33rd instruction $= N_2 \left\lfloor \frac{N_1 T_{ins}}{N_2 T_{INS} + T_{CHK}} \right\rfloor$

$T_{poll} = 82^{*} T_{INS} + T_{33} + (67 - KN_2) T_{INS}$